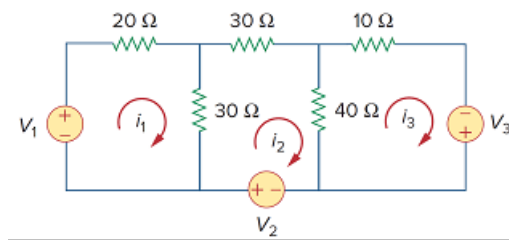


Problem

Using Fig. 3.88, design a problem to help students better understand mesh analysis using matrices.

Figure 3.88:



Step-by-step solution

Step 1 of 8

Refer to Figure 3.88 in the textbook.

Write the Kirchhoff's voltage law for loop 1.

$$-V_1 + 20i_1 + 30(i_1 - i_2) = 0$$

$$-V_1 + 20i_1 + 30i_1 - 30i_2 = 0$$

$$-V_1 + 50i_1 - 30i_2 = 0$$

$$50i_1 - 30i_2 = V_1 \dots\dots (1)$$

Step 2 of 8

Write the Kirchhoff's voltage law for loop 2.

$$30(i_2 - i_1) + 30i_2 + 40(i_2 - i_3) - V_2 = 0$$

$$30i_2 - 30i_1 + 30i_2 + 40i_2 - 40i_3 - V_2 = 0$$

$$-30i_1 + 100i_2 - 40i_3 = V_2 \dots\dots (2)$$

Step 3 of 8

Write the Kirchhoff's voltage law for loop 3.

$$40(i_3 - i_2) + 10i_3 - V_3 = 0$$

$$40i_3 - 40i_2 + 10i_3 - V_3 = 0$$

$$-40i_2 + 50i_3 - V_3 = 0$$

$$-40i_2 + 50i_3 = V_3 \dots\dots (3)$$

Step 4 of 8

Write the equations (1), (2) and (3) in matrix form.

$$\begin{bmatrix} 50 & -30 & 0 \\ -30 & 100 & -40 \\ 0 & -40 & 50 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix}$$

Write the expressions for the currents i_1, i_2 and i_3 as,

$$i_1 = \frac{\Delta_1}{\Delta} \dots\dots (4)$$

$$i_2 = \frac{\Delta_2}{\Delta} \dots\dots (5)$$

$$i_3 = \frac{\Delta_3}{\Delta} \dots\dots (6)$$

Step 5 of 8

Determine Δ .

$$\begin{aligned} \Delta &= \begin{vmatrix} 50 & -30 & 0 \\ -30 & 100 & -40 \\ 0 & -40 & 50 \end{vmatrix} \\ &= 50(5000 - 1600) + 30(-1500) + 0 \\ &= 125000 \end{aligned}$$

Determine Δ_1 .

$$\begin{aligned} \Delta_1 &= \begin{vmatrix} V_1 & -30 & 0 \\ V_2 & 100 & -40 \\ V_3 & -40 & 50 \end{vmatrix} \\ &= V_1(5000 - 1600) + 30(50V_2 + 40V_3) + 0 \\ &= 3400V_1 + 1500V_2 + 1200V_3 \end{aligned}$$

Determine Δ_2 .

$$\begin{aligned} \Delta_2 &= \begin{vmatrix} 50 & V_1 & 0 \\ -30 & V_2 & -40 \\ 0 & V_3 & 50 \end{vmatrix} \\ &= 50(50V_2 + 40V_3) - V_1(-1500) + 0 \\ &= 1500V_1 + 2500V_2 + 2000V_3 \end{aligned}$$

Determine Δ_3 .

$$\begin{aligned} \Delta_3 &= \begin{vmatrix} 5 & -30 & V_1 \\ -30 & 100 & V_2 \\ 0 & -40 & V_3 \end{vmatrix} \\ &= 50(100V_3 + 40V_2) + 30(-30V_3 + 0) + V_1(1200 + 0) \\ &= 1200V_1 + 2000V_2 + 4100V_3 \end{aligned}$$

Step 6 of 8

Determine the current i_1 .

Substitute the value of Δ , and Δ_1 in equation (4).

$$i_1 = \frac{3400V_1 + 1500V_2 + 1200V_3}{125000}$$

Determine the current i_2 .

Substitute the value of Δ , and Δ_2 in equation (5).

$$i_2 = \frac{1500V_1 + 2500V_2 + 2000V_3}{125000}$$

Step 7 of 8

Determine the current i_3 .

Substitute the value of Δ , and Δ_3 in equation (6).

$$i_3 = \frac{1200V_1 + 2000V_2 + 4100V_3}{125000}$$

Step 8 of 8

Therefore, the currents i_1 , i_2 , and i_3 are,

$$\begin{array}{l} i_1 = \frac{3400V_1 + 1500V_2 + 1200V_3}{125000} \\ i_2 = \frac{1500V_1 + 2500V_2 + 2000V_3}{125000} \\ i_3 = \frac{1200V_1 + 2000V_2 + 4100V_3}{125000} \end{array} .$$

Hence, the problem is solved according to mesh analysis using matrices.